

UNCHANGED BY THE SWAP

robodog_control_node
loop, safety, gait,
balance, estimator

robodog_web_server
GUI protocol

robodog_camera_node
publishes RGB-D

RViz2 / rosbag2
same topics

JointBackend

abstract: configure/enable/read/write/step

KinematicBackend
ideal 2nd-order joints
no physics, runs in CI

MujocoBackend
contact, armature,
1 ms command delay

CameraBackend

abstract: configure/capture/intrinsics

SimCameraBackend
MuJoCo render + z^2
range noise

RobStride02Backend

2 x CAN @ 1 Mbit/s
python-can, 400 Hz

RS02 frame codec
pure, 29 unit tests

NuwaHP60CBackend

UVC / vendor SDK
USB 3.0

Per-joint mapping
direction +1 / -1
offset_rad
bus, motor_id

The ONLY place motor
coordinates differ from
canonical joint coordinates.

Set by
calibrate_joint